

Exploring the Virtual Continuum to tackle Parkinson's Disease: AR vs VR in Therapeutic Gaming

Mariana Cachapa* João Viana† Pedro Albuquerque Santos‡ Patrícia Macedo§ Rui Neves Madeira¶

NOVA LINCS, NOVA School of Science & Technology, NOVA University Lisbon; Portugal
RESILIENCE, Polytechnic University of Setúbal, Portugal

ABSTRACT

The effective management of Parkinson's disease (PD) is increasingly reliant on innovative approaches to motivate patients to maintain physical activity. The potential of augmented reality (AR) and virtual reality (VR) as immersive tools to enhance patient engagement and encourage movement has been demonstrated. This study examines a fruit-catching game in both AR and VR formats, exploring the virtual continuum to prove which version is more beneficial for therapeutic use in tackling PD. The results of this study inform the future development of more personalized, accessible, and impactful serious games tailored to support PD rehabilitation.

Index Terms: Virtual Continuum—AR—VR—Parkinson's Disease—Serious Games—User Study.

1 INTRODUCTION AND BACKGROUND

Neurodegenerative disorders (ND), such as Alzheimer's and Parkinson's diseases (PD), progressively impair the nervous system, leading to declines in cognitive, motor, and emotional functions that significantly affect quality of life [4]. Current treatment options for ND primarily concentrate on symptom management and slowing down the disease's progression [5], involving medication, physical therapy, and cognitive interventions. The absence of a cure highlights the urgent need for engaging therapeutic strategies to enhance patient motivation and improve overall well-being.

Gamification, particularly in the context of ND, has demonstrated the potential to enhance motivation and engagement in therapy by incorporating elements of game design such as points, rewards, and challenges [1, 7]. Serious Games (SG) for therapeutic purposes can provide an interactive platform to address physical, cognitive, and psychosocial needs in an engaging manner [3].

Among the various technological advances that have the potential to improve healthcare, AR and VR stand out as promising tools, particularly within the context of SG. Virtual reality typically immerses the user in a completely synthetic environment, providing a high degree of presence and control over the virtual experience. However, this high level of immersion can lead to sensory overload, particularly in those with cognitive and motor impairments, potentially increasing the risk of disorientation, falls, and anxiety, especially in the context of ND [2].

On the other hand, AR overlays digital elements in the real world, allowing for a greater sense of familiarity and reducing the potential for discomfort and disorientation [6]. The ability of AR to seamlessly blend virtual and real offers a more accessible and flexible platform, potentially accommodating a broader range of individuals,

including those with diverse cognitive and physical capabilities [2].

This paper explores the potential of both AR and VR technologies in the therapeutic treatment of individuals with PD. By examining these modalities along the virtual continuum, the study aims to identify which approach—AR or VR—offers greater benefits for therapeutic applications.

2 USER STUDY

This study utilized a game titled *Organizing the Fruits* to compare the usability and efficacy of VR and AR for therapeutic purposes in individuals with PD. The original game was developed using Unity to improve motivation through VR in physiotherapy for individuals with PD. The game incorporates elements designed to engage players both physically and cognitively. For instance, players engage in arm extension and trunk rotation movements, replicating exercises used in conventional physiotherapy, while also incorporating elements such as color-coded fruits and baskets, directional arrows, and hand indicators to stimulate attention, memory, and decision-making skills. The *Organizing the Fruits* game simulates aerobic exercises in which players use controllers or their own hands to: 1) catch virtual fruits flying toward them from various directions, and 2) place them in appropriate colored boxes (Figure 1).

To compare the suitability of AR and VR for PD therapy, the VR version was adapted to create a comparable AR version using the pass-through projection of headsets such as the Meta Quest 3 and the Meta Quest Pro. Both versions maintained the same core gameplay mechanics (catching and sorting fruits), obviously differing in their level of immersion and the inclusion of certain virtual elements. The VR version immerses the player in a completely virtual environment with background elements. For instance, it provides virtual hands for a more realistic interaction instead of using controllers, background elements like grass and sky to enhance immersion and a virtual therapist to provide encouragement. In contrast, the AR version is less immersive, overlaying virtual elements (fruits, baskets) onto the player's real-world surroundings.

The study assessed usability, cognitive load, sense of presence, and perceived difficulty through validated questionnaires (SUS, NASA-TLX, Paas Scale, and IPQ), alongside eye-tracking data to examine attention and cognitive load. The study included a control group of 31 healthy participants (aged 18–50 years, $M = 21.8, SD = 5.84$; predominantly university students or young professionals, 24 males and 7 females) and 5 individuals with PD ($M = 65.0, SD = 6.78$, three males and two females). While the limited size of the PD group represents a constraint, the study provides important preliminary insights. Participants engaged in two-minute gameplay sessions with both AR and VR versions, with the order counterbalanced to minimize bias (see Figure 2). The games were tested using the Meta Quest Pro headset.

3 KEY FINDINGS

The NASA-TLX revealed that individuals with PD reported higher levels of mental demand in both AR and VR settings compared to the control group ($p < .05$ in several paired t-tests), indicating that

*e-mail: mariana.cachapa@estudantes.ips.pt

†e-mail: jmd.viana@campus.fct.unl.pt

‡e-mail: pedro.albuquerque@estsetubal.ips.pt

§e-mail: patricia.macedo@estsetubal.ips.pt

¶e-mail: rui.madeira@estsetubal.ips.pt

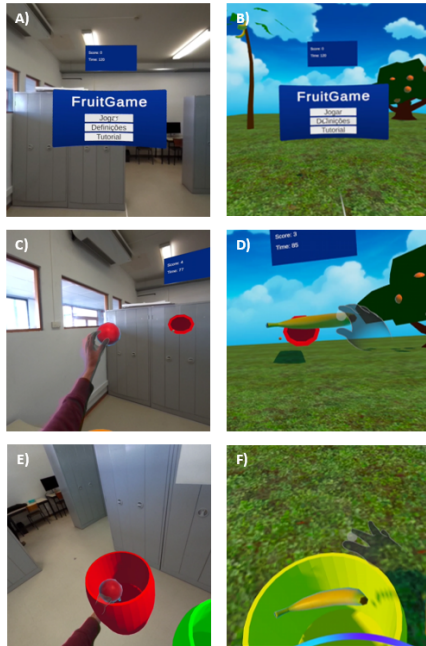


Figure 1: A) AR scenario; B) VR scenario; C) Catching a fruit (AR); D) Catching a fruit (VR); E) Putting a fruit in the basket (AR); F) Putting a fruit in the basket (VR).

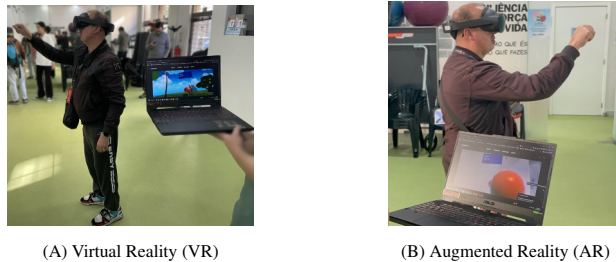


Figure 2: Participant playing both versions of the game.

the PD group experienced a greater cognitive burden as expected. This was corroborated by the **Paas Scale**, where individuals with PD rated the cognitive load higher in both environments. These findings highlight that participants with PD found both technologies more mentally demanding, likely due to the challenges of managing motor symptoms and processing visual information.

The **SUS scores**, which assess usability, indicated no significant differences between AR and VR for either group, suggesting that both technologies were similarly easy to use. However, the PD group reported higher levels of frustration and effort, particularly in the VR environment, consistent with the increased cognitive load observed in this setting. This suggests that VR might be more taxing for individuals with PD, potentially due to its immersive nature.

The **sense of presence**, as measured by the IPQ, revealed that the control group felt more immersed in the VR environment (IPQ score of 3.84) compared to AR (IPQ score of 3.12), with significant differences in the feeling of "being there." However, individuals with PD reported similar levels of presence in both AR (IPQ score of 2.83) and VR (IPQ score of 2.73), suggesting that intense sensory input in VR might be overwhelming for PD patients, reducing their sense of immersion. This observation highlights the

potential of AR as a more comfortable and less cognitively taxing environment for individuals with PD.

Eye-tracking metrics provided valuable insights into cognitive processing and attention allocation. Significant differences between the PD and control groups were observed in both AR and VR settings. In terms of fixation duration, individuals with PD exhibited longer fixation durations, particularly in VR ($p = .0020$), indicating slower processing speed or difficulty disengaging attention. Similarly, higher fixation counts in the PD group ($p = .0012$ in AR; $p = .0029$ in VR) suggest challenges in efficiently scanning the visual field. **Saccade data** also revealed significant differences, with individuals with PD showing longer saccade durations ($p = .0001$ in VR; $p = .0000$ in AR) and more frequent saccades ($p = .0006$ in AR; $p = .0029$ in VR), indicating motor control difficulties and challenges in transitioning between visual points of interest. These metrics underscore the distinct cognitive and motor processing patterns in individuals with PD, which might affect their interaction with immersive technologies.

The study examined the impact of **PD stages** on user perceptions, revealing a significant difference in perceived physical demand between Stage 1 and Stage 2 participants, particularly in VR ($p = .0198$). Individuals in Stage 2 reported higher demands, suggesting that as PD advances, VR's immersive nature becomes more challenging due to worsening motor symptoms or increased sensitivity to sensory overload.

4 CONCLUSIONS AND FUTURE WORK

This study highlights the potential of AR and VR in therapeutic gaming for PD, emphasizing tailored technology for individual needs and disease stages. AR's safety and accessibility make it especially promising for engagement and rehabilitation adherence. Larger, long-term studies are essential for effectiveness assessment to validate these findings and refine immersive therapies along the Virtual Continuum.

ACKNOWLEDGMENTS

We would like to thank Adrian Vulpe-Grigorasi (adrian.vulpe-grigorasi@fhstp.ac.at) for the eye-tracking analysis. This work is supported by UID/04516/NOVA Laboratory for Computer Science and Informatics (NOVA LINCIS) with the financial support of FCT/IP.

REFERENCES

- [1] S. Al-Rayes et al. Gaming elements, applications, and challenges of gamification in healthcare. *Informatics in Medicine Unlocked*, 31:100974, 2022. doi: 10.1016/j.imu.2022.100974 1
- [2] H. A. G. d. O. Araújo et al. Immediate effect of augmented reality, virtual reality, and neurofunctional physiotherapy on postural control and executive function of individuals with parkinson's disease. *Games for Health Journal*, 12(3):211–219, 2023. PMID: 35972381. doi: 10.1089/g4h.2021.0222 1
- [3] R. Damaševičius, R. Maskeliūnas, and T. Blažauskas. Serious games and gamification in healthcare: A meta-review. *Information*, 14(2), 2023. doi: 10.3390/info14020105 1
- [4] J. Dumurgier and C. Tzourio. Epidemiology of neurological diseases in older adults. *Revue Neurologique*, 176(9):642–648, 2020. doi: 10.1016/j.neurol.2020.01.356 1
- [5] D. S. Knopman et al. Alzheimer disease. *Nature Reviews Disease Primers*, 7(1):33, May 2021. doi: 10.1038/s41572-021-00269-y 1
- [6] R. Skarbez, M. Smith, and M. C. Whitton. Revisiting milgram and kishino's reality-virtuality continuum. *Frontiers in Virtual Reality*, 2, 2021. doi: 10.3389/frvir.2021.647997 1
- [7] V. Wattanasoontorn, I. Boada, R. García, and M. Sbert. Serious games for health. *Entertainment Computing*, 4(4):231–247, 2013. doi: 10.1016/j.entcom.2013.09.002 1